



**BHASKARACHARYA NATIONAL INSTITUTE FOR SPACE
APPLICATIONS AND GEO-INFORMATICS**

WEEKLY PROGRESS REPORT (22/06/2026 – 28/06/2026)

WEEK 6

PROJECT NAME

ROAD CONDITION DETECTION

PROJECT DESCRIPTION :

**AUTOMATIC DETECTION AND
SEGMENTATION OF ROAD DAMAGES**

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22/06/2026	Analyzed segmentation outputs and identified issues related to improper mask alignment. Investigated causes of inaccurate localization in road damage detection results.
23/06/2026	Improved the detection pipeline to achieve better alignment between predicted masks and actual road damage regions. Performed testing on multiple videos to validate the improvements.
24/06/2026	Started implementing JSON file generation for storing detection results in a structured format. Designed the JSON schema to include labels, confidence scores, and frame information.
25/06/2026	Initiated dashboard development and integrated core components for visualizing detection results. Verified data flow between the inference pipeline and dashboard modules.
26/06/2026	Holiday
27/06/2026	Holiday
28/06/2026	Holiday (Sunday)

Next Week Plan	• Complete JSON generation integration with the inference pipeline. • Enhance dashboard UI and add advanced analytics and visualization features. • Further improve segmentation accuracy and optimize real-time performance.
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REFERENCE:

1. Jocher, G., et al., *Ultralytics YOLO Documentation*. Available: <https://docs.ultralytics.com/>
2. Long, J., Shelhamer, E., & Darrell, T., "Fully Convolutional Networks for Semantic Segmentation," *CVPR*, 2015.
3. Ronneberger, O., Fischer, P., & Brox, T., "U-Net: Convolutional Networks for Biomedical Image Segmentation," *MICCAI*, 2015.
4. The Python Software Foundation, *Python Documentation*. Available: <https://docs.python.org/3/>

SCREEN SHOTS:



